Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2021-P1-Model)** **Date: 22-04-2026**

Time: 09.00Am to 12.00Pm **GTA-26** **Max. Marks: 180**

KEY SHEET

PHYSICS

1	**A**	2	**B**	3	**C**	4	**C**	5	**2**	6	**0.50**
7	**1.73**	8	**3**	9	**0.50**	10	**2**	11	**ABD**	12	**AC**
13	**ACD**	14	**BD**	15	**BD**	16	**AC**	17	**4**	18	**7**
19	**2**										

CHEMISTRY

20	**C**	21	**B**	22	**A**	23	**B**	24	4	25	94
26	**505**	27	**95**	28	**29 to 30**	29	3	30	**AD**	31	**BC**
32	**BC**	33	**CD**	34	**BCD**	35	A	36	**16**	37	28
38	**5**										

MATHEMATICS

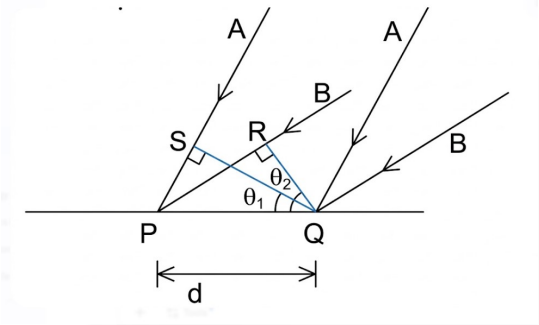
39	**B**	40	**C**	41	**B**	42	**B**	43	**6**	44	**6.75**
45	**18**	46	**12**	47	**5**	48	**49**	49	**ACD**	50	**ACD**
51	**ABCD**	52	**ACD**	53	**ABCD**	54	**AC**	55	**200**	56	**19**
57	**38**										

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SOLUTIONS

PHYSICS

1.

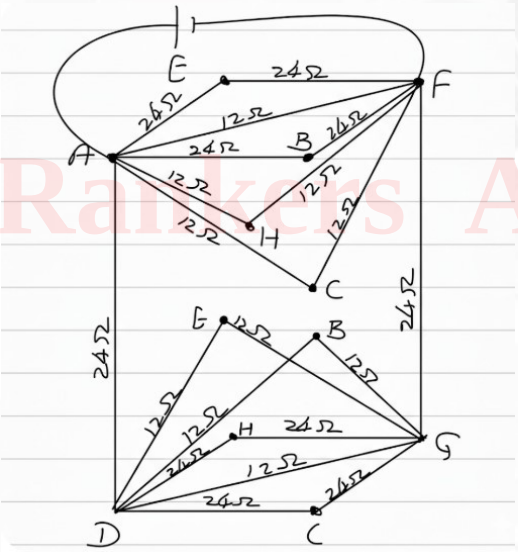


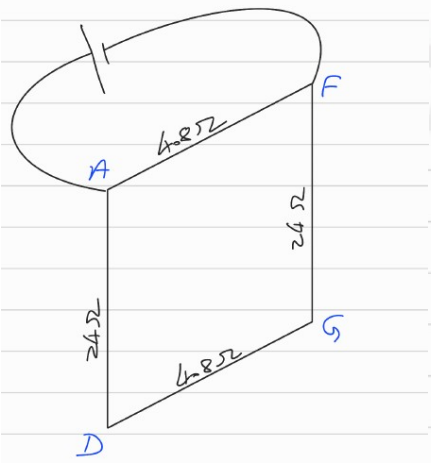
$PR = d\sin\theta_2,\ PS = d\sin\theta_1,\ \Delta x = \eta\lambda + PR - PS$

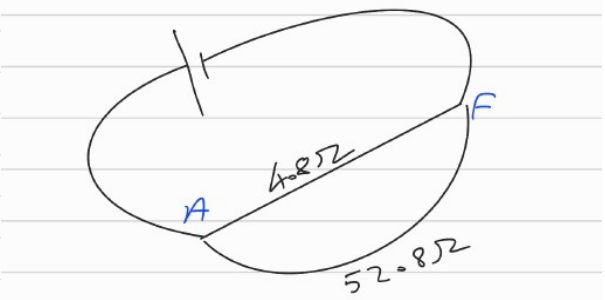
For P to be next maxima

$\Delta x = (n+1)\lambda$

2.





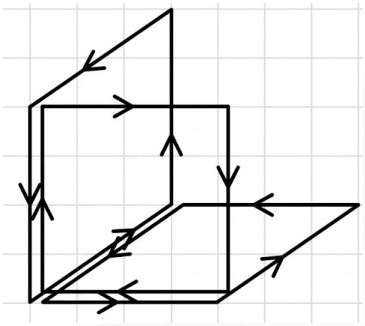


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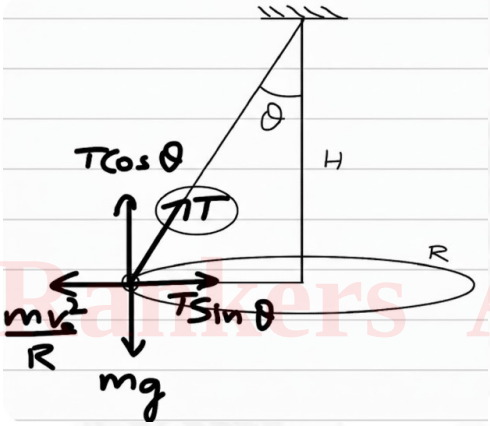
$$R_{eq} = \frac{52.8\times 4.8}{52.8+4.8} = \frac{52.8\times 4.8}{57.6}$$

$$R_{eq} = 4.4\Omega$$

3.



4.



Conserving angular momentum along vertical direction

$$mv_0R = mvr \Rightarrow r = \frac{v_0R}{v}$$

Work Done in Raising Height

W = Increase in Potential Energy + Increase in Kinetic Energy

$$W = mg\frac{H}{2} + \frac{1}{2}mv^2 - \frac{1}{2}mv_o^2$$

5. Change in momentum of each photon $= \frac{h}{\lambda}(\hat{j}-\hat{i})$

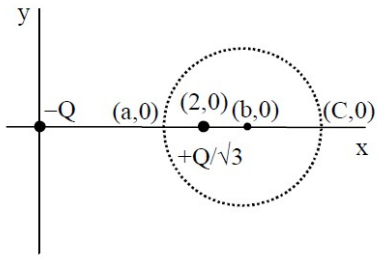
Force on mirror $F = \frac{n_p h}{\lambda}(\hat{i}-\hat{j})$

$$F = \frac{P}{C}\left(\hat{i}-\hat{j}\right)$$

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6. $E = \dfrac{Q}{2A\epsilon_0} = \dfrac{\eta P \lambda t e}{2A\epsilon_0 hc}$

7. **1.73**



Lets take two points (a, 0) and (C, 0) on equipotential circle. Net potential at (C, 0) = 0

$$\frac{K(-q)}{C} + \frac{Kq}{\frac{\sqrt{3}}{(C-2)}} = 0$$

$$\frac{1}{C} = \frac{1}{\sqrt{3}(C-2)}$$

$$\sqrt{3}(C-2) = C$$

$$\Rightarrow \sqrt{3}C - 2\sqrt{3} = C$$

$$\Rightarrow (\sqrt{3}-1)C = 2\sqrt{3}$$

$$\Rightarrow C = \frac{2\sqrt{3}}{\sqrt{3}-1}$$

Potential net at (a, 0) = 0

$$\frac{K(-q)}{a} + \frac{K\frac{q}{\sqrt{3}}}{(2-a)} = 0$$

$$\Rightarrow \frac{1}{a} = \frac{1}{\sqrt{3}(2-a)}$$

$$\Rightarrow 2\sqrt{3} - \sqrt{3}a = a$$

$$\Rightarrow a = \frac{2\sqrt{3}}{1+\sqrt{3}}$$

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$$\text{So, Radius} = \frac{C-a}{2} = \frac{\frac{2\sqrt{3}}{\sqrt{3}-1} - \frac{2\sqrt{3}}{\sqrt{3}+1}}{2}$$

$$= \sqrt{3}\left(\frac{1}{\sqrt{3}-1} - \frac{1}{\sqrt{3}+1}\right)$$

$$= \sqrt{3}\left(\frac{(\sqrt{3}+1)-(\sqrt{3}-1)}{3-1}\right)$$

$$= \sqrt{3}\left(\frac{2}{2}\right)$$

$$\text{Radius} = \sqrt{3}$$

8. **3.00**

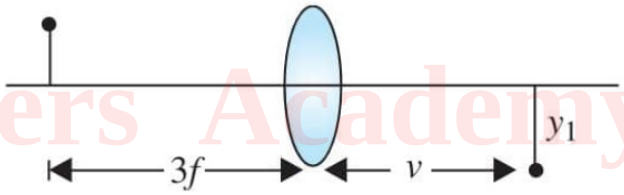
$$b = a + \text{radius}$$

$$= \frac{2\sqrt{3}}{\sqrt{3}+1} + \sqrt{3} = \frac{2\sqrt{3}+3+\sqrt{3}}{1+\sqrt{3}}$$

$$= \frac{3\sqrt{3}+3}{1+\sqrt{3}} = 3$$

Center $= (3,\ 0)$

9 & 10.



$$\frac{1}{v} - \frac{1}{-3f} = \frac{1}{f} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{3f} \Rightarrow v = \frac{3f}{2}$$

$$\frac{y_1}{y} = \frac{v}{3f} \Rightarrow y_1 = y\frac{v}{3f} = \frac{y}{2}$$

$$If\ y_{max} = A, then\ y_{1max} = A_1 = A/2$$

$$\text{Max velocity} = \omega A_1 = \omega A/2$$

11. $-\frac{dx}{dt} = 0.1\sqrt{x}$

$$\int_9^0 \frac{dx}{\sqrt{x}} = -0.1\int_0^\tau dt \Rightarrow \tau = 60 \text{ sec}$$

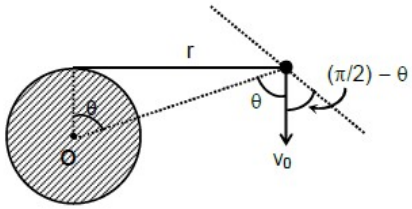
$$\omega = \frac{0.1\sqrt{x}}{x} = \frac{0.1}{\sqrt{x}} \text{ rad/s}$$

$$a = v\omega = 0.1\sqrt{x} \cdot \frac{0.1}{\sqrt{x}} = 0.01 \text{ m/s}^2$$

12. Speed of the object will remain constant because tension is always perpendicular to the velocity.

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So angular momentum about $O = mv_0 \sin\theta \frac{r}{\sin\theta} = mv_0 r$



$O = mv_0 \sin\theta \frac{r}{\sin\theta} = mv_0 r$

13. In both situations $v_C = \frac{J}{m}$

In situation-(ii), $\omega = \frac{J}{m\ell}$

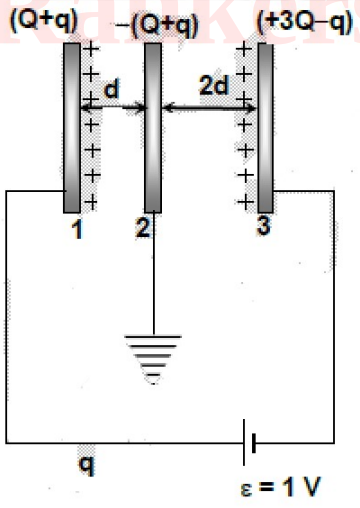
14. $H_B^{max} > H_A^{max} > H_C^{max}$ (as shown clearly)

$So, u_B^y > u_A^y > u_C^y$

Vertical component of initial velocity of each particle will decide the time taken.

So, $t_C < t_A < t_B$

15. Charge flown through S_1 *is* $Q + 3Q = 18$ mC



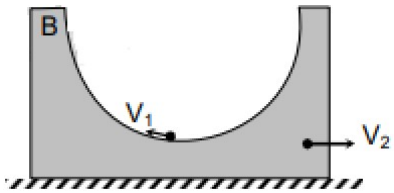
$0 + \frac{(Q+q)}{\varepsilon_0 A} d - \varepsilon - \frac{(3Q-q)2d}{\varepsilon_0 A} = 0$

By solving $q = 15$ mC

16. Using conservation of energy and momentum when particle leaves end B, velocity of the block will be zero.

At lower most point vertical acceleration is $\frac{v^2}{R}$

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$\therefore 5v_1 = 10v_2$

and $5 \times g \times 6 = \frac{1}{2} \times 5v_1^2 + \frac{1}{2} \times 10 \times v_2^2$

Solving equations $v_1 = \sqrt{80}, \quad v_2 = \sqrt{20}$

$v_{P,B} = \sqrt{80} + \sqrt{20}$

$a_{P,B} = \frac{v_{P,B}^2}{R} = \frac{(\sqrt{80} + \sqrt{20})^2}{2}$

$\vec{a}_{P,g} = \vec{a}_{P,B} + \vec{a}_{B,g}$

$\vec{a}_{B,g} = 0$ (when particle is at lowest position)

$\therefore \vec{a}_{P,g} = \vec{a}_{P,B} = 90 \text{ m/s}^2$

17. Standing Wave $y = 2A\sin(kx)\cos(\omega t)$

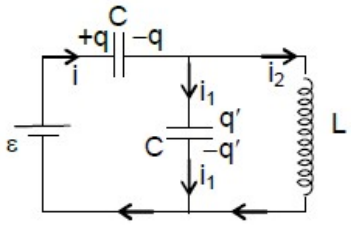
Has energy $E = \mu\omega^2 A^2 \ell$

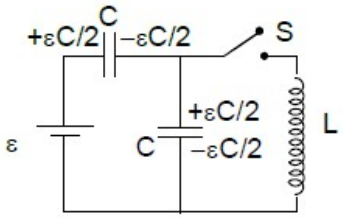
18. Using Kirchhoff's Law

$\varepsilon - \frac{q}{C} - \frac{q'}{C} = 0$

$-\frac{q}{C} + L\frac{di_2}{dt} = 0 \quad \ldots(ii)$

$\varepsilon - \frac{q}{C} - L\frac{di_2}{dt} = 0 \quad \ldots(iii)$





Solving these equations with initial conditions at

$t = 0, i_2 = 0$

$i_2 = \varepsilon\sqrt{\frac{C}{2L}}\sin\omega t$

$\therefore i_{2(\max)} = \varepsilon\sqrt{\frac{C}{2L}} = 7\text{A}$

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19. Let v be the speed of projectile at the highest point. By conservation of angular momentum and conservation of mechanical energy, we have

$$muR\sin\alpha = mvr_{max}$$

$$and\ \frac{1}{2}mu^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{r_{max}}$$

Solving these two equations, we get $r_{max} = \frac{3R}{2}$

$$\therefore\quad h_{max} = r_{max} - R = \frac{R}{2}$$

CHEMISTRY

20. ALL ARE CORRECT

21.

$\Delta T_f = i.Kf.m$

$i = 1.1$

$i = 1 - \alpha + n\alpha$

$\alpha = 0.1[\because n = 2]$

$K_a = \frac{c\alpha 2}{1-\alpha}$

$K_n = \frac{K_w}{K_a} = 9\times10^{-12}$

22. $0.5353 = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{0.1}{0.001}$

$E^\circ_{cell} = 0.5353 + 0.0591 = 0.5944 \simeq 0.59\text{V}$

$E^\circ_{cell} = \frac{0.0591}{2}\log K_c$

$K_c = 10^{\left[\frac{0.59\times2}{0.0591}\right]}$ For Forward reaction

$0.5353 = E^\circ_{cell} - \frac{0.0591}{2}\log\frac{[Ni^{2+}]}{[Cu^{2+}]}$

$0.5353 = 0.59 - \frac{0.0591}{2}\log\frac{0.1}{x}$

$x = 0.001\text{ M}$

$\text{New EMF} = 0.5353 - 0.02955 = 0.5058\text{V}$

$[Cu^{+2}]_{new} = \frac{0.001}{k}\quad [H^+]_{new} = \frac{0.01}{k}$

$0.5058 = 0.59 - \frac{0.0591}{2}\log\frac{0.1}{\frac{0.001}{k}}$

$0.0842 = 0.0295\log(100k)$

$\log(100k) = \frac{0.0842}{0.0295} \simeq 3$

$100k = 1000 \Rightarrow k = 10$

$\therefore [H^+]_{new} = \frac{0.01}{10} = 0.001$

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$$A \rightarrow B + C + \ldots\ldots Z$$

$P_i \quad 0 + 0 + \ldots 0 \quad Initially$

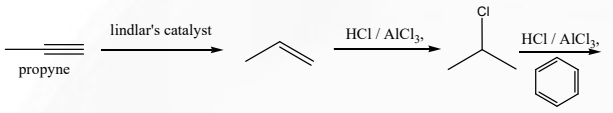
$P_i - x \quad 25x \quad At\ equilibrium$

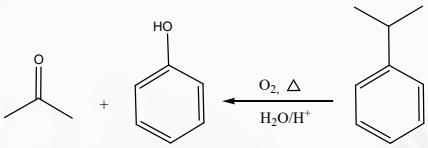
$$P_t = (P_i - x) + 25x = P_i + 24x$$

$$x = \frac{P_t - P_i}{24}$$

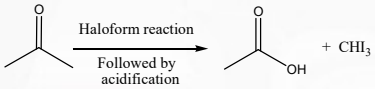
$$K = \frac{2.303}{t} \log \frac{P_t \times 24}{2P_i - P_t}$$

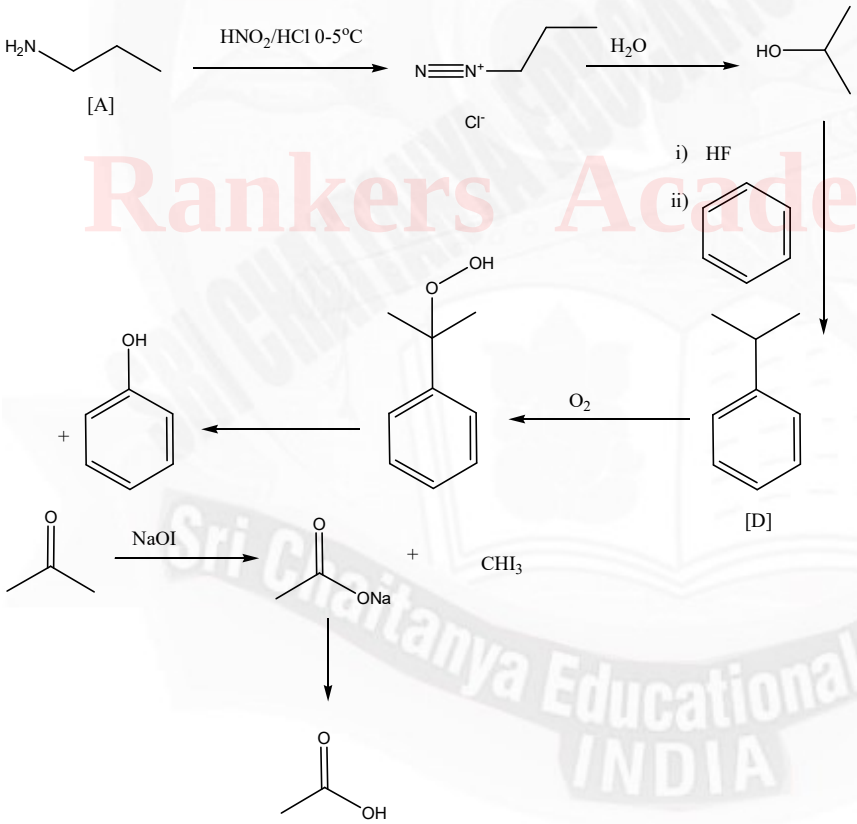
23.





24. & 25.





molecular weight of acetic acid is 60.

26. Both are 0.1 M solutions therefore react completely, so heat released in 5.75kJ but for dissociation of 0.1M ammonium hydroxide 0.7kJ is consumed therefore heat energy released is 5.75-0.7=5.05kJ=505×10^{-2}kJ

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27. $\Delta H=2\times(-57.5)+20=-95kJ$

28. $\frac{P^o-P_s}{P^o}=\frac{loss\ in\ mass\ of\ solvent\ bulbs}{total\ gain\ in\ mass\ of\ guard\ tubes}$

$P_o-P\ \alpha\ 0.01$

$P_0\ \alpha\ 0.24$

$\frac{0.01}{0.24}=\frac{W_s/180}{W_w/18}$

$29.4\%\approx 29.5\%$

$\frac{(225\times10^{-x})-(0.1944)}{225\times10^{-x}}=\frac{1}{25}\times3.4$

$(225\times10^{-x})-(0.1944)=30.6\times10^{-x}$

29. $(194.4)\cdot10^{-x}=0.1944$

$10^{-x}=10^{-3}$

x = 3

30. A and B) Molecule that forms stable carbocation undergo SN^1 reaction easily.
C and D) electron with drawing group stabilize transition state in SN^2 reaction and easy is SN^2 reaction.

31. CONCEPTUAL

32. B, C

A) Process of precipitating colloidal solution by using an electrolyte is called "COAGULATION" and not Peptisation.

B) Since, molar mass of sol is much higher than true solutions, so magnitude of any colligative properties

Is smaller than true solutions. $(\Delta T_f)_{sol}<(\Delta T_f)_{true\ solution}$

So, freezing point of sols > freezing point of true solution.

So, option (B) is correct.

(C) Micells are formed greater than or equal to CMC and above KRAFT temperature.

So option (C) is also correct.

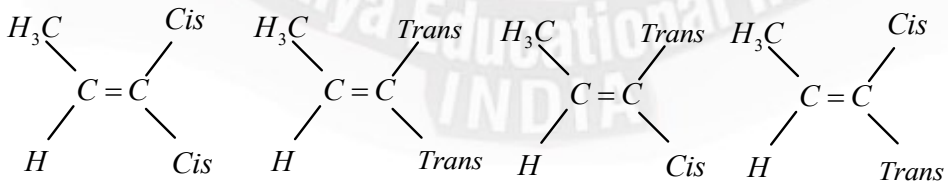
(D) Micelles are ASSOCIATED colloids and not Macromolecular colloids.

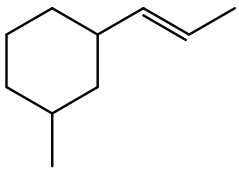
33. CONCEPTUAL

34. CONCEPTUAL

35. yellow ppt lead chromate and ppt formed in copper hydroxide.

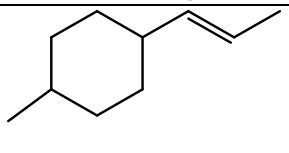
36.

i) 

ii) 

$2geo\times2^2\ optical=8\ stereo\ isomers$

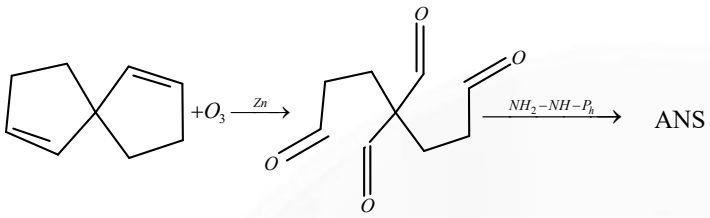
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iii) 

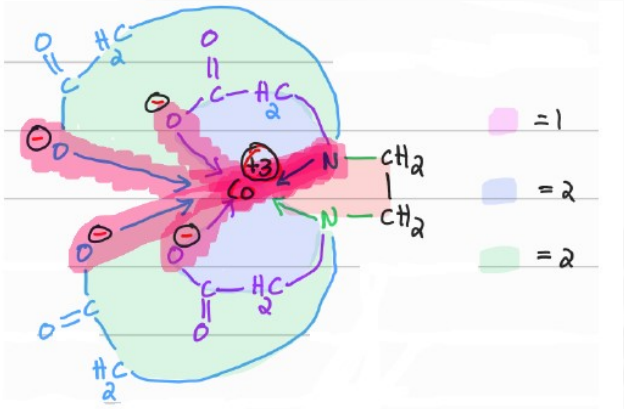
$2geo$

$Total\ stereo\ isomers = 4+8+2=14$

37.



38. $Co^{3+} + \text{EDTA}^{4-} \rightarrow [Co(\text{EDTA})]^-$



Number of cyclic rings = denticity -1 = 6-1=5.

MATHEMATICS

39. $$\lim_{n\to\infty}\left(\sum_{r=1}^{n}\frac{n^2}{n^2+r^2}\left(\lim_{m\to\infty}\sum_{k=1}^{mr}\left(\frac{1}{n^2+\frac{k^2}{m^2}}\right)\frac{1}{m}\right)\right)$$

$$=\lim_{n\to\infty}\left(\sum_{r=1}^{n}\frac{n^2}{n^2+r^2}\int_0^r\frac{1}{n^2+x^2}dx\right)$$

$$=\lim_{n\to\infty}\left(\sum_{r=1}^{n}\frac{n}{n^2+r^2}\tan^{-1}\left(\frac{r}{n}\right)\right)$$

$$=\int_0^1\frac{\tan^{-1}x}{1+x^2}dx=\frac{\pi^2}{32}$$

40. EQUATION OF TANGENT PARABOLA IS $ty = x + t^2$

It is passing through P(6,5)

$t = 2,3$

$\therefore Q(4,4)\,R(9,6)$

Equation of circle touching in the parabol at Q (4,4) and R

$(x-4)^2+(y-4)^2+\mu(x-2y+4)=0$

$(x-9)^2+(y-6)^2+\lambda(x-3y+5)=0$, these are passing through focus (1, 0)

$r_2 = 5\sqrt{10}$

$r_1 = \frac{5}{2}\sqrt{5}$

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41. $As\ x \to 0, \tan^m x \approx x^m .So\ f(x) \approx \dfrac{x^n \cos(1/x)}{x^m} = x^{n-m}\cos(1/x)$

$$f'(0) = \lim_{h\to 0}\frac{f(h)-f(0)}{h} = \lim_{h\to 0}\frac{h^{n-m}\cos(1/h)}{h} = \lim_{h\to 0} h^{n-m-1}\cos(1/h)$$

42. $\mathrm{P}\left(\dfrac{\text{luggage was lost at Dubai}}{\text{luggage does not reach Muscat}}\right)$

$$\frac{\left(1-\frac{1}{20}\right)\left(1-\frac{1}{10}\right)\frac{1}{5}}{1-\left(1-\frac{1}{20}\right)\left(1-\frac{1}{10}\right)\left(1-\frac{1}{5}\right)\left(1-\frac{1}{10}\right)} = \frac{855}{1922}$$

43. $ABC \perp DBC \Rightarrow (\overrightarrow{AB}\times\overrightarrow{AC})\cdot(\overrightarrow{DB}\times\overrightarrow{DC}) = 0$

$$(\vec{b}\times\vec{c})\cdot((\vec{b}-\vec{d})\times(\vec{c}-\vec{d})) = 0$$

$$\Rightarrow (\vec{b}\times\vec{c})\cdot(\vec{b}\times\vec{c}-\vec{b}\times\vec{d}-\vec{d}\times\vec{c}) = 0$$

$$\Rightarrow |\vec{b}\times\vec{c}|^2 - (\vec{b}\times\vec{c})\cdot(\vec{b}\times\vec{d}) - (\vec{b}\times\vec{c})\cdot(\vec{d}\times\vec{c}) = 0$$

$$\Rightarrow |\vec{b}|^2|\vec{c}|^2\cdot\frac{3}{4} - \begin{vmatrix}\vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{d}\\ \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{d}\end{vmatrix} - \begin{vmatrix}\vec{b}\cdot\vec{d} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{d} & \vec{c}\cdot\vec{c}\end{vmatrix} = 0$$

$$\Rightarrow \frac{3|\vec{b}|^2|\vec{c}|^2}{4} - |\vec{b}|^3|\vec{d}|\begin{vmatrix}1 & \frac{1}{4}\\ \frac{1}{2} & \frac{1}{2}\end{vmatrix} - |\vec{b}|^3|\vec{d}|\begin{vmatrix}\frac{1}{4} & \frac{1}{2}\\ \frac{1}{2} & 1\end{vmatrix} = 0$$

$$\Rightarrow \frac{3|\vec{b}|}{4} - 6\left(\frac{1}{2}-\frac{1}{8}\right) - 6\left(\frac{1}{4}-\frac{1}{4}\right) = 0$$

$$\Rightarrow \frac{3|\vec{b}|}{4} = \frac{9}{4}$$

$$\Rightarrow |\vec{b}| = 3 \Rightarrow |\vec{c}| = 3 \Rightarrow |\vec{b}| + |\vec{c}| = 6$$

44. Volume of tetrahedron

$$= \frac{1}{6}\left[\vec{d}\ \vec{b}\ \vec{c}\right] = \frac{27}{4}$$

$$(z-2)^8 = 3^8$$

45. $\left(\dfrac{z-2}{3}\right)^8 = 1$

$z_1, z_2, \ldots, z_8$ Lies on the circle with centre (2, 0) and radius =3

Area of quadrilateral is 18

46. $(z-2)^8 = 3^8$

$$\left(\frac{z-2}{3}\right)^8 = 1$$

$z_1, z_2, \ldots, z_8$ Lies on the circle with centre (2, 0) and radius =3

$= 6$

47. The points of intersection of internal and external angle bisectors of

$B, C\ are\ I(-2,-3), I_1(-6,-2) and\ B\left(-\frac{3}{2}, \frac{5}{2}\right).$

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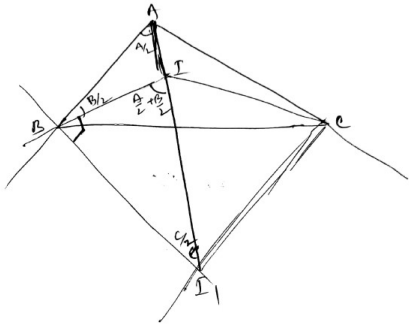
I_1, B, I_1, C *are concyclic.*

Circumcentre of ΔABC *is mid point of* $I_1I_1 = \left(-4, \frac{1}{2}\right)$

$\angle BI_1I = \frac{C}{2}$

$\tan\frac{C}{2} = \frac{1}{9}$

$\tan C = \frac{9}{40}$



48.

$m_{BC} = \frac{13}{6}$

$\angle BI_1I = \frac{C}{2}$

$\tan(C/2) = \left|\frac{3-13/6}{1+3(13/6)}\right| = \left|\frac{18/6-13/6}{1+39/6}\right| = \left|\frac{5/6}{45/6}\right| = \frac{5}{45} = \frac{1}{9}$

$\tan C = \frac{9}{40}$

49. Solution Since $|z_1| = |z_2| = |z_3|$ Circumcentre triangle ABC at the origin.

Orthocentre z_p of triangle ABC is $z_1 + z_2 + z_3$.

Also angle subtended by AB and orthocentre are A + B and A + C respectively.

⇒ angles subtended by the sides AB and AC at the orthocentre are equal.

$$\arg\left(\frac{z_p - z_1}{z_p - z_3}\right) = \arg\left(\frac{z_p - z_2}{z_p - z_1}\right)$$

$$\arg\left(\frac{z_2 + z_3}{z_1 + z_2}\right) = \arg\left(\frac{z_1 + z_3}{z_2 + z_3}\right)$$

$$\arg\left(\frac{(z_2 + z_3)^2}{(z_1 + z_2)(z_1 + z_3)}\right) = 0$$

$\frac{(z_1 + z_3)(z_1 + z_2)}{(z_2 + z_3)^2}$ is purely real.

50. BY SIMPLYFYING

$y^3 = 2x - 3y$

$2x = y^3 + 3y \Rightarrow x = \frac{y^3 + 3y}{2}$

the inverse function is $f^{-1}(n) = \frac{n^3 + 3n}{2}$

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51. Solution: Let $g(x)=[f(x)]^m\cdot[f(1-x)]^n, where\ m,n>0$

$g(0)=g(1)=0$

$\therefore$ By Rolle's Theorem, there exists some $c\in(0,1)$ such that $g'(c)=0$

$g(x)=f(x)\cdot f^n(1-x)$

Applying the derivative

$-f(x)\cdot n\cdot f^{n-1}(1-x)\cdot f'(1-x)+f'(x)\cdot f^n(1-x)=0$

$f^{n-1}(1-x)\left[f'(x)f(1-x)-f(x)\cdot n\cdot f'(1-x)\right]=0$

$\frac{f'(x)}{f(x)}=n\frac{f'(1-x)}{f(1-x)}$

52. $C=(A-B)(A+B)(A-B)^{-1}$

$(A-B)(A+B)=(A+B)(A-B).$

$C=(A+B)(A-B)(A-B)^{-1}$

$C=(A+B)I$

$C=A+B$

53. $(x-y)f(x+y)-(x+y)f(x-y)=2y(x-y)(x+y)$

$Let\ x-y=u,x+y=v\Rightarrow uf(v)-vf(u)=(v-u)uv\Rightarrow\frac{f(v)}{v}-v=\frac{f(u)}{u}-u=k$

$\Rightarrow f(x)=kx+x^2\ \because f(1)=2\Rightarrow k=1\Rightarrow f(x)=x^2+x$

54. Foci of ellipse $S_1\ (1,1)\ \ S_2\ (4,5)$

$a=\frac{\sqrt{5}\left(2\sqrt{2}+1\right)}{2}$

$b=\sqrt{5\left(\sqrt{2}+1\right)}$

55. $2000x^6+\underbrace{100x^5+10x^3+x}_{\text{a G.P.}}-2=0$

$2000x^6+\frac{x((10x^2)^3-1)}{10x^2-1}-2=0$

$\frac{x(1000x^6-1)}{10x^2-1}=-2(1000x^6-1)$

$1000x^6-1=0$ or $\frac{x}{10x^2-1}=-2\ \Rightarrow x=-\left(10x^2-1\right)$

$\Rightarrow x^2=\frac{1}{10}$ Which is not possible $20x^2+x-2=0$

$x=\frac{-1\pm\sqrt{1+160}}{40}$

$x=\frac{-1+\sqrt{161}}{40}$ or $\frac{-1-\sqrt{161}}{40}$

$m=-1;n=161;r=40$

$m+n+r=-1+161+40=200$

56. Consider 4 positive terms

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$$\frac{r_1}{2},\frac{r_2}{4},\frac{r_3}{5},\frac{r_4}{8}$$

$$\text{A.M.}=\frac{1}{4}\left(\frac{r_1}{2}+\frac{r_2}{4}+\frac{r_3}{5}+\frac{r_4}{8}\right)=\frac{1}{4}\times 1=\frac{1}{4}$$

$$\text{G.M.}=\left(\frac{r_1}{2}\cdot\frac{r_2}{4}\cdot\frac{r_3}{5}\cdot\frac{r_4}{8}\right)^{1/4}=\left(\frac{r_1\cdot r_2\cdot r_3\cdot r_4}{2\cdot 4\cdot 5\cdot 8}\right)^{1/4}$$

$$r_1r_2r_3r_4=\frac{5}{4}$$

$$\therefore \text{G.M.}=\left[\frac{5}{4(2\cdot 4\cdot 5\cdot 8)}\right]^{1/4}=\left(\frac{1}{2^8}\right)^{1/4}=\frac{1}{4}$$

Hence, **A.M. = G.M.**

All numbers are equal:

$$\frac{r}{2}=\frac{r_2}{4}=\frac{r_3}{5}=\frac{r_4}{8}=k$$

$$r_1=2k;\quad r_2=4k;\quad r_3=5k;\quad r_4=8k$$

$$\Rightarrow\prod r_i=(2\cdot 4\cdot 5\cdot 8)k^4\quad \frac{5}{4}=(2\cdot 4\cdot 5\cdot 8)k^4$$

$$\therefore k=1/4$$

Hence:

$$r_1=\frac{1}{2};\quad r_2=1;\quad r_3=\frac{5}{4};\quad r_4=2$$

$$\Rightarrow\sum r_i=\frac{19}{4}\quad \text{but}\quad r_1+r_2+r_3+r_4=\frac{a}{4}$$

$$\Rightarrow\frac{19}{4}=\frac{a}{4}\quad\Rightarrow \mathbf{a=19}$$

57. $f(n,r)=\dfrac{n!}{r!(n-r)!}\times\dfrac{r!3!}{(r+3)!}$

$$=\frac{n!3!}{(n-r)!(r+3)!}$$

$$=\frac{n!3!}{(n+3)!}\times\frac{(n+3)!}{(n-r)!(r+3)!}$$

$$\frac{n!3!}{(n+3)!}\times{}^{n+3}C_{r+3}$$

$$g(n)=\sum_{r=0}^{n}(-1)^r\frac{n!3!}{(n+3)!}\,{}^{n+3}C_{r+3}$$

$$=\frac{n!3!}{(n+3)!}\left[(-1)^{0n+3}C_3+(-1)^{1n+3}C_4+\ldots+(-1)^{nn+3}C_{n+3}\right]$$

$$=\frac{n!3!}{(n+3)!}\left(\frac{(n+1)(n+2)}{2}\right)$$

$$g(n)=\frac{3}{n+3}$$

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